AUTOMATIC GATE CONTROL FOR SMART PARKING SYSTEM USING CIRCULAR QUEUE AND MINHEAP

PROJECT REPORT

Submitted By

MOUNIDHARAN V	2022504003
PREM ARUN P	2022504012
GUNA S	2022504007
THIRULOGASUNDAR S	2022504309

In partial fullfillment for the award of the degree
of

BACHELOR OF ENGINEERING

IN

ELECTRONICS AND COMMUNICATION ENGINEERING

DEPARTMENT OF ELECTRONICS ENGINEERING

MADRAS INSTITUTE OF TECHNOLOGY

ANNA UNIVERSITY CHENNAI -600 044

APRIL 2025

ANNA UNIVERSITY: CHENNAI 600 04

BONAFIDE CERTIFICATE

Certified that this project report" **AUTOMATIC GATE CONTROL FOR SMART PARKING SYSTEM USING CIRCULAR QUEUE AND MINHEAP"** is the Bonafide work of **"MOUNIDHARAN V(2022504003),PREM ARUN P(2022504012), GUNA S (2022504007),THIRULOGASUNDAR S(2022504309)"**who carried out the project work under my supervision.

SIGNATURE

Dr. RADHA SENTHILKUMAR

SUPERVISOR

Associate Professor,

Department of Information technology,

Madras Institute of Technology,

Anna University,

Chennai – 600044.

ACKNOWLEDGEMENT

We consider it as our privilege and our primary duty to express our gratitude and respect to all those who guided and inspired us in the successful completion of the project.

With profound gratitude, we sincerely acknowledge with thanks the opportunity provided to us by our respectful **Dr.K.RAVICHANDRAN**, Dean, Madras Institute of technology, Anna University for providing good environment and facilities.

We thank our respectful Head of the Department **Dr.P.T.V.BHUVANESHWARI,** Professor and Head of Electronics Engineering, Madras Institute of technology, Anna University for his encouragement during the Project work.

We sincerely express our gratitude to our guide **Dr.RADHA SENTHILKUMAR,** Associate Professor, Department of Electronics Engineering for her valuable guidance throughout the course of this project.

Place: Chennai

Date:05/05/2025

MOUNIDHARAN V-(2022504003)

PREM ARUN P-(2022504012)

GUNA S-(2022504007)

THIRULOGASUNDAR S -(2022504309)

ABSTRACT

This project introduces a smart parking system that automates vehicle entry and parking slot allocation using the Arduino Uno, ultrasonic sensors, and a servo motor. The goal is to address common problems in conventional parking systems such as human dependency, time delays, and inefficient space utilization. The system employs a queue data structure to manage the orderly entry of vehicles and a min heap algorithm to dynamically allocate the nearest available slot to each vehicle.

Ultrasonic sensors are used to detect the presence of vehicles both at the entry point and within individual parking slots. When a vehicle is detected at the entrance, and a slot is available, it is added to the queue. The nearest free slot is then determined using the min heap, and the servo motor is triggered to open the gate, allowing the car to enter. Once a car vacates its slot, the slot is marked available and reinserted into the heap.

This project successfully demonstrates how simple hardware components and efficient algorithms can be integrated to build a low-cost, real-time, intelligent parking management system. It serves as a foundation for more complex IoT-based smart city infrastructure and can be further expanded with features like app integration and real-time monitoring.

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1.INTRODUCTION

BACKGROUND

Urban areas around the world are facing increasing challenges related to vehicle parking due to rapid urbanization and rising vehicle ownership. Traditional parking systems, which rely on manual supervision and allocation, often result in long wait times, inefficient use of available space, and increased fuel consumption due to cars idling or searching for free slots. This not only leads to traffic congestion around parking facilities but also contributes to environmental pollution and user frustration.

In many locations, parking areas do not have mechanisms to monitor occupancy in real time, and users have no way of knowing which slots are available without physically checking. This lack of real-time data and automation creates inefficiencies in both public and private parking zones.

To address these issues, the development of smart parking systems is becoming increasingly important. By leveraging sensors, microcontrollers, and algorithms, these systems aim to optimize space utilization, reduce human intervention, and improve user convenience. This project contributes to that effort by implementing an automated parking system that uses data structures and sensors to streamline the parking process.

OBJECTIVES

- Automate gate control based on slot availability.
- Assign the nearest available parking slot using an efficient algorithm.
- Detect car presence and update system status in real time.

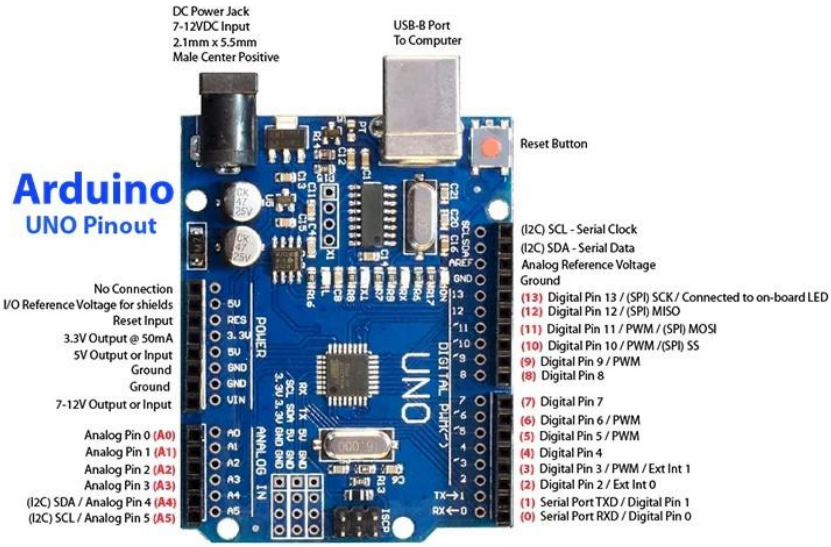
PROBLEM STATEMENT

Manual parking leads to congestion and inefficient space usage. A system is needed that can detect free slots and control vehicle entry without human intervention.

2.METHODOLOGY

TOOLS USED

1. ARDUINO UNO



Fig(i): Arduino UNO pinout

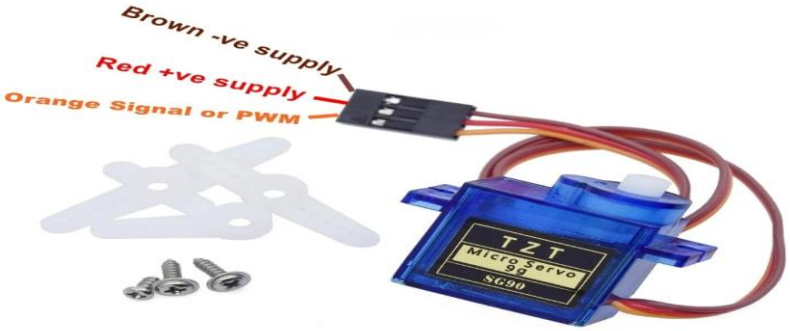
The Arduino Uno is a microcontroller board based on the ATmega328P. It is widely used in embedded systems and DIY electronics projects due to its simplicity and flexibility. It has 14 digital input/output pins, 6 analog inputs, a USB connection for programming, and a power jack. In this project, it serves as the central controller, handling sensor inputs, motor outputs, and implementing logic for queue and min heap operations.

2.HC-SR04 ULTRASONIC SENSORS

Fig(ii):HC-SR04 pinout

- The HC-SR04 is a popular ultrasonic distance sensor used to measure the distance to an object by sending out ultrasonic waves and measuring the time it takes for the echo to return. It consists of a transmitter and a receiver and works effectively within a range of 2 cm to 400 cm.
- In this project, these sensors are used to:
    - Detect vehicle presence at the entry point.
    - Monitor occupancy status of parking slots.

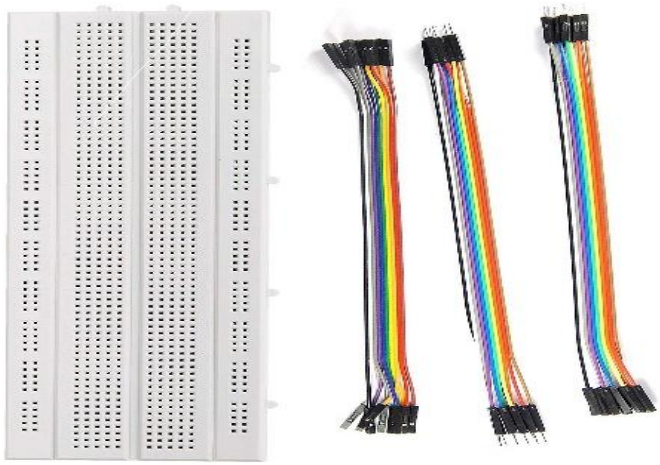
3.SERVO MOTOR



Fig(iii):HG-servo motor

- A servo motor is a rotary actuator that allows precise control of angular position.
- It includes a motor, a control circuit, and a feedback system.
- In this project, a standard servo motor (typically SG90 or similar) is used to simulate the gate mechanism.
- It opens the gate when a vehicle is allowed to enter and closes it afterward

4.JUMPER WIRES AND BREADBOARD



Fig(iv):Breadboard and Jumper wires repectively

- Jumper wires are used to make connections between different components on a breadboard without soldering.
- A breadboard is a construction base for prototyping electronics. It allows users to build and test circuits easily.
- These tools are essential for assembling the circuit in a non-permanent way for development and testing.

DESIGN

HC-SR04 ULTRASONIC SENSORS

- HC-SR04 ultrasonic sensors are used to detect incoming vehicles. It measures the distance to the nearest object in front of it. If this distance falls below a defined threshold (e.g., 7 cm), the system identifies a vehicle attempting to enter.
- Slot Sensors: Installed in each parking slot to determine whether a slot is occupied or available. If a vehicle is parked in a slot, the sensor's measured distance becomes very short, indicating occupancy.
- This non-contact method of detection is cost-effective and reliable for short-range applications like smart parking.

SERVO MOTOR

A standard servo motor is used to simulate the automated gate at the parking entry:

- The servo is programmed to rotate between 0° (closed) and 90° (open).
- When a vehicle is allowed to enter (i.e., after slot assignment), the servo opens the gate.
- After a brief delay (allowing the vehicle to pass), the gate is closed again.
- This automated gate control adds a layer of realism and security to the system, mimicking actual automated parking infrastructure.

CIRCULAR QUEUE

- **VEHICLE ENTRY HANDLING**
  - When a vehicle is detected at the entry gate, it is assigned a unique identifier and added to the rear of the circular queue if space is available.
- **FIFO LOGIC WITH WRAP-AROUND:**
  - The circular queue follows the First-In, First-Out (FIFO) principle, meaning vehicles are parked in the order they arrive. When the rear index reaches the end of the array, it wraps around to the beginning, making full use of the array space.
- **OVERFLOW & UNDERFLOW PREVENTION:**
  - The design includes checks to prevent overflow (when the queue is full) and underflow (when the queue is empty). This ensures reliable operation even under rapid vehicle movement.
- **WAITING MANAGEMENT:**
  - If all parking slots are occupied, vehicles remain in the queue until a slot becomes available. Once a slot frees up, the next vehicle in line is dequeued and assigned a slot.
- **PERFORMANCE:**
  - The circular queue offers O(1) time complexity for both enqueue and dequeue operations, making it highly efficient for real-time applications like smart parking systems.

MIN HEAP

Parking slot management is optimized using a min heap, a binary tree structure where the smallest element is always at the root:

- Each slot is assigned an index based on its physical distance from the gate.
- Initially, all slots are inserted into the heap to indicate availability.
- When a vehicle is ready to be parked, the system extracts the minimum (i.e., the nearest available slot) from the heap.
- Once the vehicle exits the slot, that slot index is reinserted into the heap, updating the system's state dynamically.
- This ensures quick access to the best available slot with minimal computational overhead.

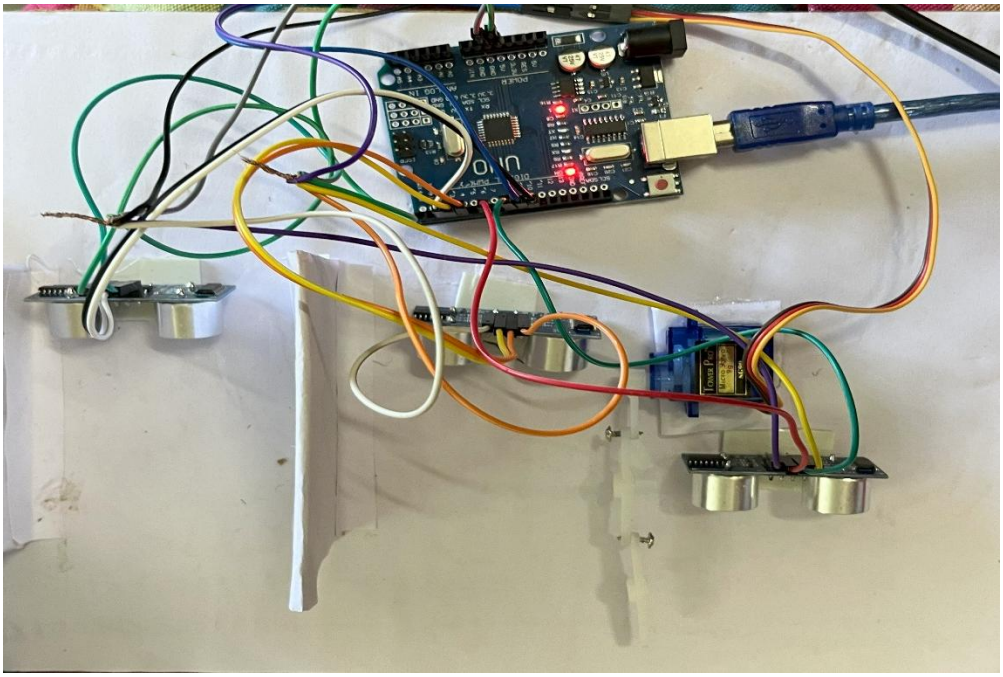
RESULTS

DESCRIPTION

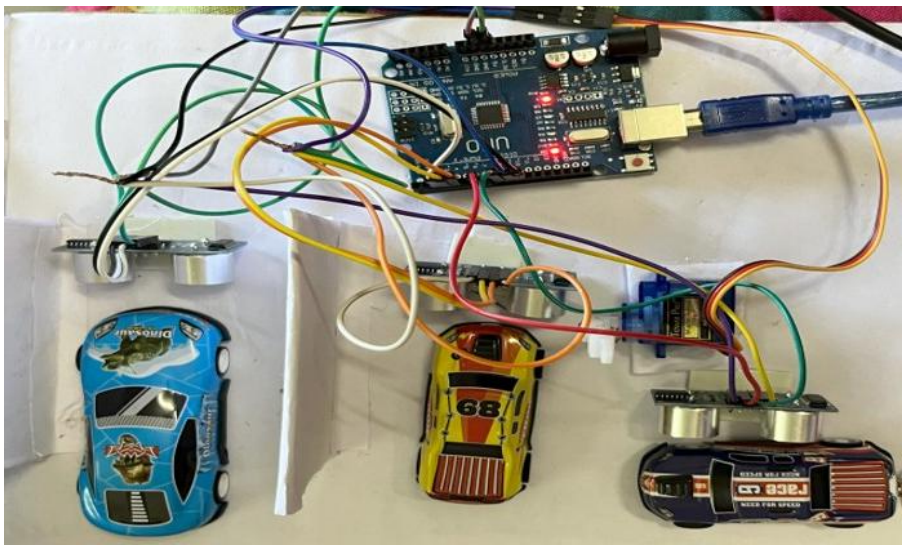
The project successfully implemented an automated gate control system for a smart parking slot using Arduino and several key components such as ultrasonic sensors, a servo motor, and data structures like a queue and min heap. The system performs the following functions:

- Real-time Vehicle Detection: Ultrasonic sensors detect vehicle presence both at the entry point and within parking slots.
- Queue Management: Vehicles are queued in FIFO order when waiting to enter the parking area.
- Min Heap for Slot Assignment: The nearest available slot is assigned to the vehicle using a min-heap data structure, ensuring efficient slot allocation.
- Gate Control: The servo motor simulates gate control, opening when a vehicle is granted entry and closing once the vehicle has passed.
- Dynamic Heap Update: The min heap is updated dynamically when a slot is freed by a vehicle exiting the parking area.

HARDWARE SETUP



Fig(v): Hardware Setup



Fig(vi): Hardware Setup along with the cars in the parking slot

SERIAL MONITOR OUTPUT

```
Slot 1: Available (16 cm)
Slot 2: Available (15 cm)
Entry Sensor: 4 cm
Available Slots: 2
-----------------------
Car 960 added to queue
 available Slot near the gate =slot 1
Gate Closed
Slot 1: Occupied (4 cm)
Slot 2: Available (15 cm)
Entry Sensor: 16 cm
Available Slots: 1
```

Fig(vii): Both slots are available to occupy

```
Slot 1: Occupied (4 cm)
Slot 2: Available (15 cm)
Entry Sensor: 6 cm
Available Slots: 1
-----------------------
Car -6870 added to queue
 available Slot near the gate =slot 2
Gate Closed
Slot 1: Occupied (4 cm)
Slot 2: Occupied (4 cm)
Entry Sensor: 44 cm
Available Slots: 0
```

Fig(viii):One slot is occupied and one slot is available to occupy

```
Slot 1: Occupied (4 cm)
Slot 2: Occupied (4 cm)
Entry Sensor: 1 cm
Available Slots: 0
-----------------------
Car -8460 added to queue
Slot 1: Occupied (4 cm)
Slot 2: Occupied (2 cm)
Entry Sensor: 259 cm
Available Slots: 0
```

Fig(xi):Both slots are occupied

CONCLUSION

The system efficiently automates vehicle detection, slot allocation, and gate control using **Min Heap** and Queue. It reduces congestion and manual effort, enhancing parking efficiency. Minor improvements in sensor accuracy and scalability can further optimize performance. Future upgrades like RFID, AI-based predictions, and cloud integration can enhance its applicability.

SUMMARY:

The project achieved real-time vehicle entry control and slot assignment using Arduino and basic electronics. It demonstrates how embedded systems and data structures can solve practical problems like smart parking.

LIMITATIONS

- Only two slots used due to hardware limitations.
- No GUI or app interface for slot display.

FUTURE WORK

- Integration with a mobile app using Blynk or Adafruit IO.
- Expandable to more slots and multiple gates.
- Add sensors for vehicle exit detection.

REFERENCES

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